

Semeia Institute: Building the Foundation for Impact Investment in Brazilian Conservation Units

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On August 13, 2013, Ana Luísa, director of the Semeia Institute, had just arrived at the São Paulo office when she received an e-mail from a foreign fund interested in investing in a public-private partnership project (PPP) to explore commercially the tourism potential of the conservation units (UCs) of the so-called Lund Route located north of Belo Horizonte, the capital of the Brazilian state of Minas Gerais. Covering 18% of the country's land mass, UCs are areas protected by Federal Law 9.985/2000 for their biodiversity, scenic beauty and cultural importance. The Semeia Institute believed that many of these areas had the potential to become major sources of social and economic development and to reduce poverty. Parks such as Iguaçu Falls, Fernando de Noronha and Chapada dos Veadeiros were some of the most famous examples of UCs. However, these received few visitors, were not well known and did not play an important part in the government plans, in contrast to other countries, such as South Africa, United States, New Zealand and Costa Rica.

Fully financed by Anima Investimentos with the aim of serving as its strategic philanthropy arm, the Semeia Institute had been created driven by the belief that the management agenda of the UCs required the inclusion of more players, especially from the private sector, which brings resources and management instruments that

¹ We thank Leticia Neves for her research support and the Latin America Impact Economy Innovations Fund (Rockefeller Foundation, Avina and Omidyar, in a proposal coordinated by ICE) for their financial support. This case study is solely for the purpose of classroom discussion and does not purport to render an opinion on managerial decisions or to serve as a primary source of data.

complement activities that are normally the responsibility of the public sector. In this way, the potential of these UCs, most of which were managed by the public sector, could be better explored through partnerships with private players that invest in activities such as tourism, which would have positive impacts on the sustainability of these areas and on local development and in turn increase the value of public assets. With this idea in mind, Semeia began to work intensely on solutions that fostered stronger relationships between state governments and UC managers on the one side and potential investors and managers from the private sector interested in administering these areas on the other.

One of the most recent cases on which Semeia was working was a potential PPP for the Lund Route, a touristic route that includes natural monuments and state parks. Consisting of a series of archeological sites and most importantly caverns created during the Cenozoic Era, the Lund Route was a UC of rare beauty as well as of historical and cultural significance. However, despite its strategic location only 30 minutes north of Belo Horizonte, Minas Gerais, the route received few visitors compared to other tourism activities in the state, such as the Inhotim park in the city of Brumadinho and the historical town of Ouro Preto. The Lund Route was also suffering from conservation issues due to human activity, with the construction of private properties in the UC areas and the recurring wildfires that have affected conservation areas.

The Minas Gerais state government, aware of the route's potential and the need to attract external financing, had launched a few weeks earlier a public consultation prior to the publication of the bid notice for the UC's public auction. The Semeia Institute had played an important role in this process by bringing international experiences and conducting, through a technical cooperation process, economic feasibility studies for the tourism activities on the route. And, within approximately 60 days after the public consultation period, the government would receive investment proposals for the private concession of the UC with duration of 30 years. With the concession contract signed, the UC would most likely be managed by a consortium of private companies that would be responsible for executing the project and for managing the unit.

In such a scenario, the foreign fund would want to understand the full potential of the project. The fund had a particular characteristic: it focused on impact investing, which are investments that seek measurable social and environmental impacts as well as financial returns. The goal of the impact fund was to help finance the consortium that would eventually take over the route's management. Although a number of issues were still pending, the timing was perfect, since the government's public consultation process aimed precisely to gather suggestions to improve the proposal prior to the launch of the bidding process. Naturally it would be expected, for instance, that impact investors would want to ensure not only the project's financial attractiveness, but also how winning social and environmental impact would be effectively pursued by the consortium. Which contractual aspects would have to be complied with between the fund and the concession administrator and between the concession administrator and

the government? What risks would be involved and how could they be mitigated to attract the interest of the fund as well as that of other potential impact investors?

The Semeia Institute

The Semeia Institute, which launched its activities in 2011, is financed by Anima Investimentos, was created as a non-governmental organization (NGO) focused on forming and promoting partnerships between the public and private sectors to conserve and develop protected areas, including the surrounding areas.² The Institute's mission was "to transform protected areas into a source of pride for all Brazilians." To spur interest by the general public in the topic, Ana Luísa would frequently explain over and over to journalists:

It is by creating jobs and generating income, increasing the number of visitors and promoting the inclusion of these areas in society's imagination that the UCs will become a source of pride for all Brazilian and consequently be maintained.

Note that Semeia did not effectively manage or invest directly in the projects for the UCs. However, its efforts sought to better align the aspirations of the public sector with the concerns of the private sector in terms of both financial and socio environmental performance. This, Semeia provided philanthropic support to projects to improve the management of the country's UCs by providing the government with data and studies, and informing private agents of the potential opportunities of impact investing.

In this sense, Semeia focused its efforts on three key initiatives: "Innovation," generating knowledge, conducting studies and developing tools to expand the information available on UCs; "Engagement/Promotion," disseminating relevant content, reporting the findings of studies and expanding public interest in the topic; and "Testing," by applying the ideas and knowledge generated in the field and narrowing the relationship between public administrators and private investors. In addition to the three pillars mentioned above, Semeia had a fourth pillar, which was to continuously improve the management of the institute through planning, exemplary execution of internal processes and investing in employees and partners in order to create the foundation for the sustainable growth of Semeia as an institute (Appendix 1). As part of the "Innovation" pillar, Semeia had identified international protection areas, mainly parks, in 14 countries that were being commercially explored through partnerships between the public and private sectors and through partnerships between the public sector and NGOs. Semeia's objective to create benchmarks on the impact of PPPs on the management of parks, in particular on environmental conservation and/or local tourism (Appendix 2). According to Semeia, the results showed that the highest results in terms of revenue and impact on local income and employment were in parks where the commercial management of tourism had been granted to the private sector, such as the cases of Fiordland National Park in New Zealand and Kruger National

² Available at: < <http://www.semeia.org.br/> >. Visited on July 30, 2013.

Park in South Africa. Meanwhile, the most successful cases in terms of environmental conservation were those in which the partnerships involved NGOs, such as in the Bonaire National Marine Park in the Dutch Caribbean and in the Central Park in the United States. These benchmarks supported Semeia in its dialogue with the public administrators of the UCs in Brazil, which is discussed in the next section.

Conservation Units in Brazil

The Semeia Institute's mission of transforming Brazil's UCs into sources of wealth was based on the country's potential in this area. By 2010, some 20% of all biodiversity on the planet were located in Brazil, and the country has around 1.5 million km² (or approximately 18% of its land mass) protected in the form of UCs. Considering only federal, state and municipal parks, which are the focus of the Semeia Institute, given their potential for commercial tourism in 2010. Brazil had a total area of 346,000 km², which is an area similar in size to Germany.

However, despite all this ecological potential, investments per hectare in UCs in Brazil were modest at an average of R\$4.43, compared to R\$110.39 in New Zealand and R\$156.12 in the United States, both of which grant concessions for their national parks. On the other hand, the number of hectares per employee was much higher than in other countries. In Brazil, each employee was responsible for monitoring an average of 18,600 hectares, compared to 2,350 hectares in New Zealand and 2,120 hectares in the United States in 2011 (Appendix 3).

In terms of legislation, Brazilian UCs were regulated by Federal Law 9.985/2000, which created the National System of Nature Conservation Units (or SNUC in the Portuguese acronym)³. Article 2, Item I of the law introduced the following definition:

I - conservation unit: part of the country's territory and its environmental resources, including jurisdictional waters, with relevant natural features, legally created by the Government, with defined conservation objectives and limits, under special administrative regime, for which adequate protection guarantees shall apply.

Note that the legislation places a strong emphasis on the issue of environmental protection and preservation. However, it does not preclude the possibility of forming partnerships or the commercial exploration of the Units for tourism purposes, as stated in Article 5, Item IV, which addresses the guidelines of the SNUC:

IV – [the SNUC] seeks to support and promote cooperation with non-governmental organizations, private organizations and individuals for the development of studies, scientific research, environmental education practices, recreational activities, ecological tourism, monitoring, maintenance and other activities involving the administration of the conservation units;

³ Available at: <http://www.planalto.gov.br/ccivil_03/leis/19985.htm>. Visited on July 30, 2013.

Despite the permission for establishing public-private partnerships to explore the UCs, particularly parks that support the commercial exploration of tourism activities, the Semeia Institute observed that 83% of the administrators of UCs said that they did not generate any revenues from their public use⁴. Only 29% of the administrators of UCs declared having formed any sort of external partnership. Furthermore, 86% were in favor of outsourcing their administration to expand the offering of tourism-related goods and services at the Units⁵.

The case of the Lund Route UC in Minas Gerais

Located around 30 minutes north of the Belo Horizonte metropolitan area in the state of Minas Gerais, the Lund Route is a UC formed by three units: the natural monuments “Rei do Mato Grotto” and “Maquiné Grotto – Peter Lund Natural Monument,” and “Sumidouro State Park,” which have 52 caverns and some 170 archeological sites (Appendix 4). The route was named after Danish archeologist and explorer Peter Wilhelm Lund (1801-1880). Given the archeological wealth found in the region's caves, Lund became a permanent resident of Lagoa Santa, Minas Gerais and explored local caves for nearly ten years, which earned him the title of “the father of Brazilian archeology.”⁶ Some of Lund's most important discoveries include evidence of humans coexisting with megafauna, such as the saber-toothed cat and the giant ground sloth. Even after Lund's death, the area continued to attract scientists and researchers. In the 1970s, fossilized human remains were found dating back 60,000 years, making them the oldest in the Americas.⁷

In addition to its historical appeal, the route also is important for its environmental aspects, given its size of over 2,000 hectares. The economic and tourism potential of the UC is also connected with its proximity to a major urban center such as Belo Horizonte. The Minas Gerais state government plans to encourage the economic development of the northern metropolitan area – where the Tancredo Neves – Confins airport is located (project “North Vector”) around 50 km from the capital and 21 km from the parks (Appendix 5). According to Rogério Tavares, the manager of Sumidouro Park:

Sumidouro Park was forgotten and for 30 years existed only on paper. However, with the construction of the Tancredo Neves – Confins Airport and later the focus on the so-called North Vector, the park gained impetus and even became an environmental prerequisite for government projects.

The government then began to analyze the possibility of forming partnerships for the unit's administration and even defined a State Structuring Program specifically for the Lund Route. The program's goal was to promote environmental conservation combined with the environmentally correct commercial exploration of state parks. The

⁴ Instituto Semeia “Uso Público e Parcerias para Conservação e Desenvolvimento,” 2012, pg. 4.

⁵ Op. cit., pg. 24.

⁶ MARCHESOTTI, A. P. LEAL, F. C. “Lago Santa e a Origem do Homem na América.”

⁷ Op. cit.

public-private partnership project, therefore, was created as a result of the inclusion of the UC in the state's "structuring projects".⁸ It was then that the Semeia Institute sought out the state government to discuss the potential of the UCs. According to Cecília Vilhena from the State Forest Institute (IEF-MG):

There were already plans [by the Minas Gerais government] to come up with an innovative model for administrating the Lund Route as a tourism product and not merely as a conservation project. (...) So Semeia sought the state government [of Minas Gerais] at a very opportune time and was able to take advantage of the mutual interests.

Semeia initially established, in November 2012, a Technical Cooperation Agreement with the Minas Gerais state government, specifically with the State Economic Development Department (SEDE) and the State Environment and Sustainable Development Department (SEMAD). In the agreement, Semeia undertook to compile the data required for developing a management model of the Lund Route through a public-private partnership. In exchange, the state government create a conditions favorable for the project's progress and also monitor the activities.

As a way to encourage governmental progress on the project, the Semeia Institute donated to the state government the information compiled by expert consultants in the economic, legal and technical-environmental fields that made it possible for the government to prepare feasibility studies. The main objective of the initiative was to increase the range of high-quality information available to the government to support the subsequent formation of the public-private partnership. The government then adjusted the reports to meet the rules established and recommendations made in its management plans with the aim of determining the maximum demand in terms of visitors at each of the three units forming the Lund Route in order to not compromise the environmental conservation of the UC. According to Cecília Vilhena from the State Forest Institute (IEF-MG):

The Semeia Institute played a fundamental role in enabling us to reach this moment with the bid notice ready by providing encouragement, contributing to the project's initial phase and demonstrating an attractive path. And it also acted as a partner in the sharing of its expertise (...) If the team had been forced to conduct the process internally without this initial information, it would have been very difficult to have made progress.

This joint effort enabled the government to prepare the bid notice for the Lund Route public-private partnership, whose public consultation process was about to be launched.

The PPP and its public consultation

⁸ "Structuring projects" of the State of Minas Gerais are those considered priorities by the state government; they have specific budgets and limited control of targets for their achievement.

Federal Law 11,079 of 2004 established rules for bid processes and the contracting of public-private partnerships by government administrations and defined this type of partnership as a type of concession.⁹ According to the legislation, more specifically its Article 2, a public-private partnership is defined as follows:

Article 2. A public-private partnership is an administrative concession contract that can be either a sponsorship concession or a management concession.

Paragraphs 1 and 2 of the same Article 2 distinguish sponsorship concessions (where the concessionaire may charge users, such as toll roads) and management concessions (where the government pays a monthly fee to the concessionaire for services rendered). The specific case of the Lund Route PPP is classified as a management concession, which includes transfers from the Minas Gerais state government (monthly public fees) and ancillary revenues from tourism-related services at the UCs.

However, the concession was accompanied by a series of responsibilities. The state government tried to ensure profitability and partially guarantee demand related the risks. Meanwhile, the consortium winning the bid process would be responsible for achieving levels of excellence in terms of environmental management, infrastructure improvements and local social development. Note that the winning consortium would be the one proposing at the auction the lowest bid for the annual consideration to be received. In other words, the winner would be the one proposing the lowest annual fee from the state. Since the payments to the concessionaire would be made by the state, it was established that the government would link the amount of the installments to the achievement of goals stipulated in the contract. Note, therefore, that the PPP agreement sought to link the fee to the concessionaire's performance based on the pre-defined metrics: Paragraph 1, Article 5 of the Public-Private Partnership Law (Federal Law 11,079/04) reads:

Paragraph 1 – The agreement may provide for the payment to the private partner of variable compensation linked to its performance, in accordance with targets and with quality and availability standards defined in the contract.

In the specific case of the Lund Route, three groups of impact indicators were established related to the following aspects: i) management of the natural, historical and cultural heritage; ii) level of visitation; and iii) relationship with local communities. Each of these groups was subdivided into a total of 18 quantitative indicators used to guide the activities of the concessionaire (Appendix 6). Based on these indicators, the concessionaire would receive a "Final Performance Score" (NF) that would be used to determine the total fee to be paid by the state (Appendix 7). Assuming a "winning fee bid" of R\$6 million/year and an average ticket per visitor of R\$100, depending on the concessionaire's performance score (NF) based on the impact indicators defined in the

⁹ Available at: http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2004/lei/111079.htm . Visited on August 6, 2013.

bid notice, the total monthly fee(PM) to be paid by the state would increase or decrease. For instance, considering an average visitation estimate of 249,000 visitors/year and (on a simplified basis) a constant visitation rate of 20,750 visitors/month, if the concessionaire had a NF of 0.4, it would have to pay the state a monthly fee of around R\$141,000; whereas with a NF of 0.9, the state would have to pay the concessionaire an approximate monthly fee of R\$108,000. In other words, the evaluation of impact indicators would determine the NF received by the concessionaire, which in turn would be used to determine the amount of governmental payment that it would receive (Appendix 8).

In addition to the Financial Performance Score (NF), the visitation rate would also determine the fee to be paid by the government. The public consultation notice indicated an estimated increase in the number of visitors to the Lund Route over the 30 years of the concession (Appendix 9). If the project turned out not to be as successful as expected, the risk would be shared by the parties. Likewise, if the project proved more successful than initially expected, the agreement provided for the possibility of sharing revenues, with the transfer of some of the earnings to the government. Based on the aforementioned parameters, suppose now the concessionaire attains the highest levels in its Final Performance Score and impact indicators (NF = 1.1), but that the number of visitors to the UCs is extremely low in the year, at around 105,000 (8,750/month). In this case, the government would pay a fee of approximately R\$477,000 per month to the concessionaire. On the other hand, if the number of visitors greatly exceeds the initial expectations of the project¹⁰, with 550,000 visitors/year (approximately 46,000/month), the concessionaire would not only cease to receive the government fee, but would have to pay the government the amount of R\$320,000/month for the earnings in excess of that estimated from the unit's commercial exploration (Appendix 10). In the words of Aline Assis Buccini, an employee of the State Economic Development Department (SEDE):

The mechanism of transfers functions as a "cup of water: if the final performance score is high, but visitation demand is low, the government would "fill the cup" of the private partner. However, as tourism demand increases, the fee decreases, until it reaches a point where visitation is so high that the "cup overflows," and the private partner starts paying the government a portion of the profits earned from the right to commercially explore the UCs. However, if the final performance score is lower than 0.25, the concessionaire would not only cease to receive any fee, but would have to pay a fine to the state, and if this situation persists for more than six months, the contract could be lawfully terminated.

The public consultation notice makes it possible to verify the project's cash flow already based on the visitation estimates (Appendix 11), which indicate that the project would have an internal rate of return considered attractive by the private sector. However, in addition to the financial and legal issues, PPP projects should also take into account the political risk involved. Changes in government could interrupt the payment of fee to concessionaires. Although there are no reports of interruptions in the

¹⁰ The projections included in the public consultation, Appendix XI, indicate an estimate of approximately 430,000 visitors in year 30 of the concession.

payment of considerations, there is, for instance, the case of the state of Rio Grande do Sul, where the government agency promoting public-private partnerships was discontinued and all projects in the portfolio were terminated after a new administration took office in 2011. To mitigate any risks, the government of Minas Gerais determined that each partnership contract would include a specific list of assets to be given as collateral should the state government fail to fulfill its role and pay the monthly fee. The objective was to reduce political risks due to changes in administrations, as well as to ensure the long-term financial return of PPP projects involving large initial private investments, as was the case with the Lund Route PPP.

Lastly, in accordance with Federal Law 11,079/04, before the signing of the contract between the government and the concessionaire (in the form of a private Special Purpose Entity created for the PPP), a public consultation must be conducted lasting at least 30 days to receive suggestions prior to the publication of the bid notice. This phase of the bid process was very important for the parties involved (community, government, executors and investors), essentially because in this phase it would be possible to gather the private sector's perceptions of the project's financial potential and impact. In the words of Thiago Ferreira, a civil servant at the State Economic Development Department (SEDE):

It is during the public consultation that the government verifies the receptivity of the market and of investors with regard to investing their resources in the PPP project.

The Impact Investment Fund

To finance its investments, the SPE should establish ways to raise own or third party funds. In Brazil, PPPs were financed with the own capital of concessionaires, complemented by loans from the Brazilian Development Bank (BNDES) at interest rates lower than those offered by private-sector banks. There was, however, the possibility of attracting various other players, including the so-called impact investors. As defined by J.P. Morgan:

Impact investments are investments intended to create positive impact beyond financial return.¹¹

Investors that are interested in this type of investment differ from traditional investors primarily because they allocate their resources to projects and activities with positive social and environmental impact. However, this movement differs also from conventional philanthropy, through which resources are donated as a grant i.e. without the need for repayment. Impact investments, on the other hand, seek to achieve measurable social and environmental impact, as well as reasonable financial return to ultimately remunerate the initial investment made. One emblematic example

¹¹ J.P. Morgan, Impact Investments An emerging assets class, 2010, pg. 7.

of impact investment is Britain's Social Impact Bonds. The pilot project that was under development involved a prison supported by impact investors. Under the model, investor profitability depended on a clearly identified social target: the percentage of inmates that returned to prison compared to the rates of other “control” prisons not supported by the investors. By promoting the effective rehabilitation and reintegration into society of the former inmates, the government could save resources on inmate maintenance costs, and these savings could be used to remunerate the investors supporting the project.

As a result, a number of impact investment funds started to seek out investment projects with clear and measurable indicators. In 2013, impact investments were experiencing frank growth. In Brazil, within two years the number of impact investment funds operating in Brazil increased from three to ten, which included funds such as VOX Capital, MOV and FIRST Brasil. There was also an expectation of the entry of more foreign funds, such as Bamboo Finance Latin America (from Colombia) and LGT (from Lichtenstein). Globally, estimates made by J.P. Morgan in 2010 indicated that the impact investment segment had the potential to grow from US\$400 million to US\$1 billion by 2020.¹²

In the case of a the usual PPP, however a potential impact investor would have to act in partnership with a group of players, including contractors, concessionaires of other parks, hotel operators, tourism operators and environmental management companies. Although not all companies should formally participate in the SPE, each could contribute complementary competencies to the management and sustainability of the business.

The impact investment fund that was particularly interested in the project was LUX Social Fund¹³, a fund formed by a group of investors from the European Community. The fund had not defined a specific financial return, but demanded the creation of impact indicators that could be accurately measured and validated. There was a global trend in this type of fund to seek out a small number of indicators of high importance to the project's social and environmental impact and preferably to compare the project to other similar initiatives not affected by the investment, which could serve as control groups for comparisons.

Back at the offices of the Semeia Institute

Responding to an e-mail from LUX Social Fund¹⁴, the fund interested in the Lund Route PPP, Ana Luísa encouraged it to participate in the public consultation

¹² Op. cit., p. 6.

¹³ “LUX Social Fund” is a fictitious name; similarly, its role in the case is merely illustrative.

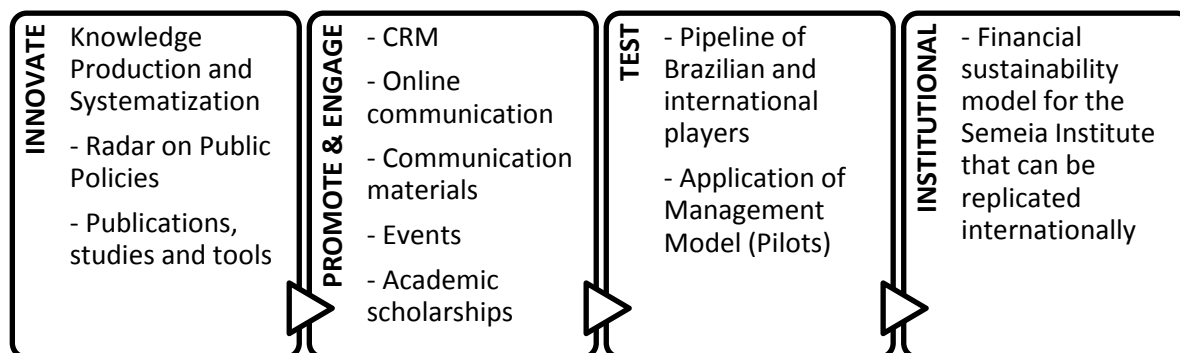
¹⁴ Idem.

to be carried by the government to explain the project in detail. Since the Semeia Institute had worked in technical cooperation with the state government of Minas Gerais, Ana Luísa limited her involvement to encouraging the fund to participate in the public consultation. She also noted that the consultation would be a good opportunity for the fund to submit to the government its suggestions for improvements to the PPP, since it could potentially serve as a benchmark in the country.

LUX Social Fund was particularly interested in understanding the quality of the impact metrics established as well as the incentives the parties would have to reconcile socio-environmental impact and financial performance. As suggested by Ana Luísa, forging a closer relationship with the government in the formal meetings of the public consultation process would also be a way to suggest improvements to the contractual arrangement and indicators of the PPP to ensure attractiveness for private investment and the attainment of socio-environmental objectives inherent to the project.

Appendices

Appendix 1 – Operating Structure of the Semeia Institute



Source: Semeia Institute Annual Report, 2012, p. 7 Extrapolated data.

Appendix 2 – Benchmarking of International Parks

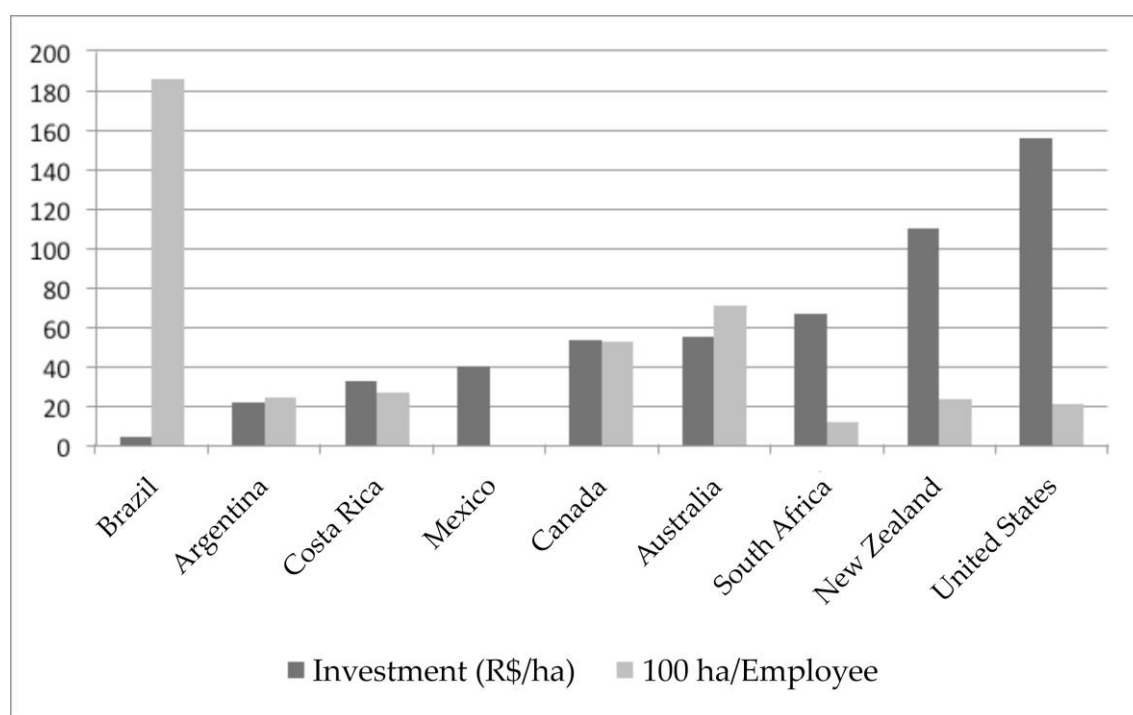


Country	Protected Area	Area Management	Tourism Services
Chile	Torres del Paine National Park	Public	Private
Argentina	Nahuel Huapi National Park	Public	Private
Colombia	Tayrona National Park	Public	Private
Caribbean Islands	Bonaire Marine National Park	Co-management	Private
Costa Rica	Volcan Poas National Park	Public	Private
Canada	Banff National Park	Public	Private
USA	Grand Teton National Park	Public	Private
USA	Central Park	Co-management	Private

Kenya	Amboseli National Park/ Area of the Olkejuado County Council	Public/ Community	Private
South Africa	Kruger National Park/ Makuleke Contract Park	Public / Community	Private
Namibia	Khoadi Hoas / Community	Private	Private
New Zealand	Fiordland National Park	Public	Private
Indonesia	Komodo National Park	Public (as of 2010)	Private (through 2010)
Italy	Cinque Terra National Park	Public	Cooperatives and Companies
Slovakia	Lower Tatra National Park	Public	N/A

Source: Semeia Institute Annual Report, 2012, p.9 Working data.

Appendix 3 – Investment per hectare and number of hectares per employee



Source: Medeiros, R.; Young; C.E.F.; Pavese, H. B. & Araújo, F. F. S. 2011. Contribution of Brazilian conservation units to the national economy: Executive Summary. Brasília: UNEP-WCMC, p.10. Working data.

Appendix 4 – Interior of Maquiné Cavern – Peter Lund Natural Monument (Cordisburgo, MG) and Interior of Rei do Mato Cavern (Lagoa Santa, MG)

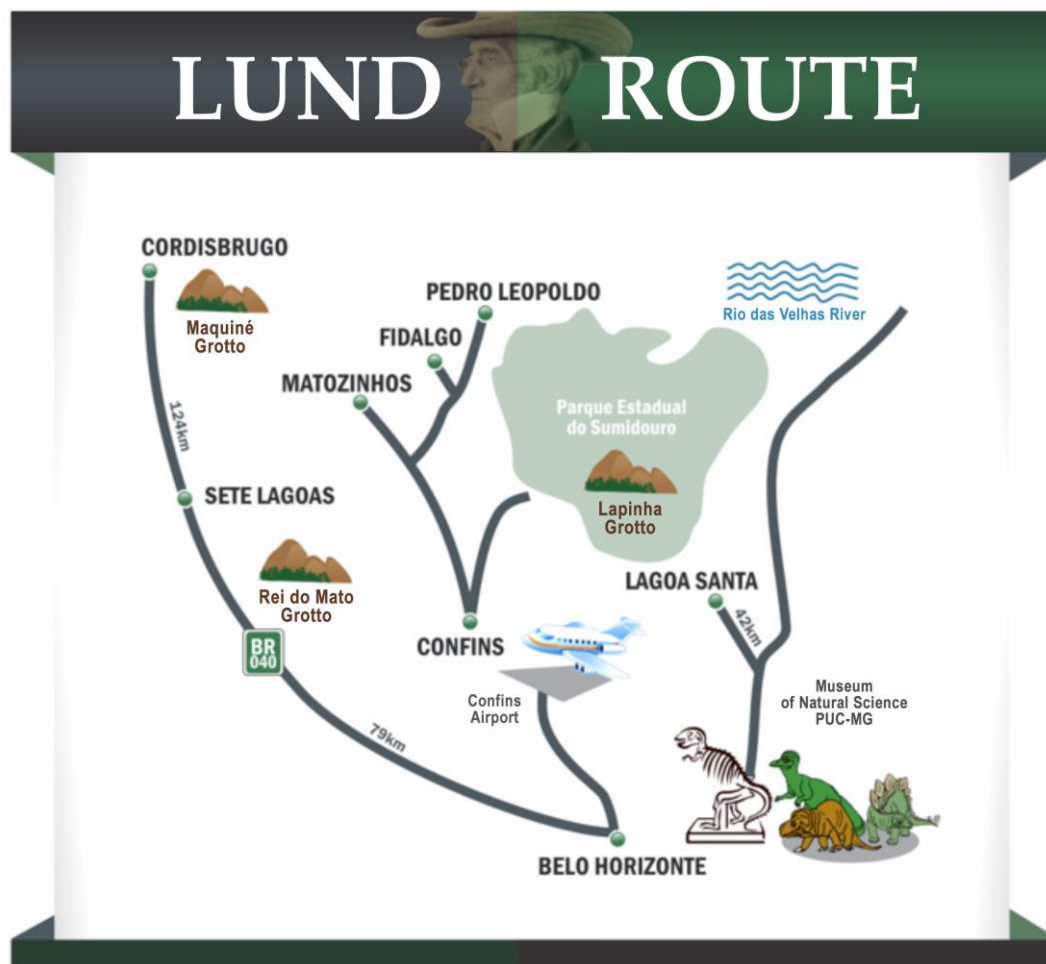


Source: Minas Gerais State Tourism Department Credit: Sérgio Mourão - Setur MG Archives. Available at <http://www.turismo.mg.gov.br/noticias/936-noticias> >. Visited on August 19, 2013.



Source: Minas Gerais State Tourism Department Setur MG. Archives. Available at <http://www.minasgerais.com.br/?s=gruta+rei+do+mato> >. Visited on August 19, 2013.

Appendix 5 – Illustrative Map of the Grotto Route



Source: Agência Minas. Available at: <http://www.agenciaminas.mg.gov.br/noticias/rota-das-grutas-peter-lund-e-uma-viagem-no-tempo/>. Visited on August 26, 2013.

Appendix 6 – Concessionaire Performance Indicators

GROUP 1. MANAGEMENT OF NATURAL, HISTORICAL AND CULTURAL HERITAGE	GROUP 2. VISITATION	GROUP 3. RELATIONSHIP WITH LOCAL COMMUNITY
Effective registration of digital security images in pre-defined priority surveillance areas	Level of satisfaction of visitor experience	Index of the inclusion of professionals with regional experience at the Conservation Unit
Maintenance index for the entire fence perimeter of Sumidouro State Park	Implementation of the safety management system for the adventure tourism activities sold by the concessionaire at the UCs	Index of environmental education events organized by the concessionaire for residents of the Buffer Zone of the 3 Conservation Units
Effectiveness and efficacy of the reporting of fire events	Percentage of visitors to more than one UC of the Lund Route	Level of satisfaction of participants of the environmental education events organized by the concessionaire with the services rendered
Maintenance of clearings to prevent forest fires	Availability of the Visitor Registration System	Level of satisfaction of users-citizens with the whistleblowing, complaint and suggestion service
Effective control of burned areas		Level of availability to citizens of whistleblowing, complaints and suggestions service
Cleaning of right-of-ways of all internal, surrounding and adjacent access roads		
Availability of updated data on environmental monitoring for the general public		
Availability of minimum protection team		
Occurrence of damage to historical, cultural and archeological heritage		

In group 1, for example, the score for the indicator “Effective registration of digital security images in pre-defined priority surveillance areas” is defined on a monthly basis based on the period during which the records remain unavailable in the system. In other words, if the images are unavailable for less than 24 hours, the concessionaire receives a score of one (1); as the period of service unavailability increases, the score also decreases gradually until service unavailability exceeds 120 hours, for which the concessionaire receives a score of zero (0).

In another case, such as in group 2, the indicator “Level of satisfaction of visitor experience,” the concessionaire must conduct a bimonthly survey of users based on a sample determined by the government that represents the diversity of users. Accordingly, if 90% or more of users are satisfied, the concessionaire receives a score of one (1), if between 90% and 85% of users are satisfied, the score is 0.7; from 85% to 70%, the score is 0.5; and if less than 70% of users are satisfied, the score is 0.2.

Note that each indicator has a specific measurement method that is predefined in the bid notice. The results obtained are later audited, in some cases by the government and in others by an independent auditor.

Source: Public consultation notice (2013). Appendix IV, pgs. 21 and 34. Available at: <<http://www.ppp.mg.gov.br/projetos-ppp/projetos-em-elaboracao/rota-lund-1/Edital%20e%20Anexos.zip/view>>. Visited on August 9, 2013.

Appendix 7 – Model of Monthly Compensation Paid to the Concessionaire

$$PM = (CP/12)*NFm - [Rm*(P+Km+Mm)+Nm*T]$$

where:

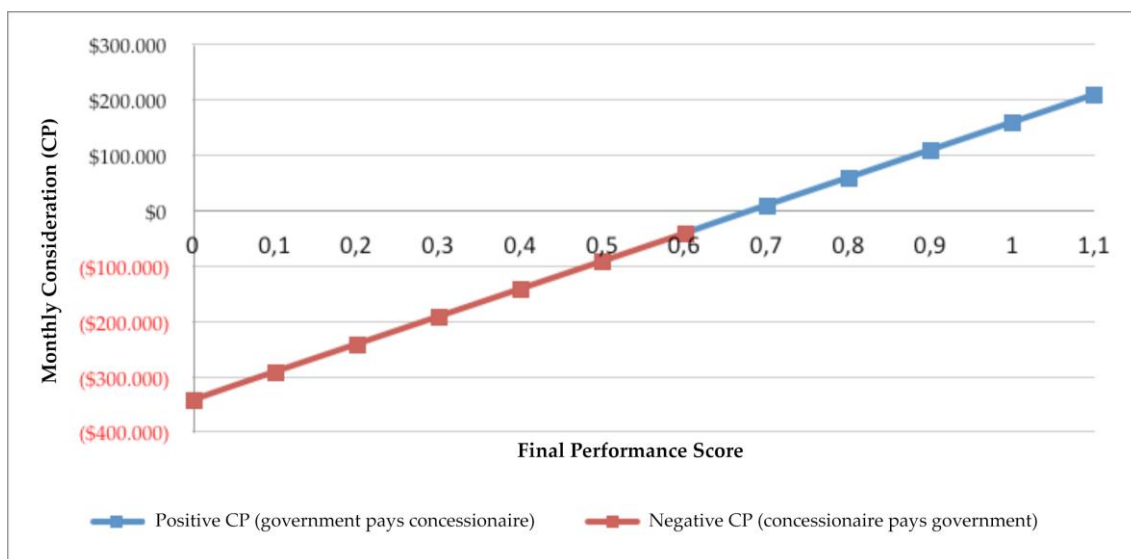
- PM = monthly cash payment;
- CP = annual fee of the winning bidder;
- Rm = revenue from ancillary activities in the period (in this case, monthly);
- R = revenue from ancillary activities in the period (in this case, annual);
- P = minimum government interest in revenue from ancillary activities:
 - P= 2%
- Km = adjustment factor as a function of the number of visitors:
 - K=0% if visitors $\leq$ (100,000/12)
 - K= Minimum [10%; 5.0%*(visitors- (100,000/12)) / (100,000/12)]
- Mm = Adjustment factor as a function of revenue:
 - M= 0% if R $\leq$ (R\$ 50 million/12)
 - M = Minimum [4%; 2%* (R -R\$ 50 million/12)/ (R\$ 50 million/12)]
- Nm = number of visitors in the period (in this case, monthly);
- T = fee paid by each visitor:
 - T = R\$ 3.00
- NFm = final score attributed to the month calculated based on the set of performance indicators described in Appendix IV – Manual of Performance Indicators of the CONTRACT

According to Appendix IV, pgs. 12-13: “The achievement by CONCESSIONAIRE of a result for a group of performance indicators of less than twenty-five percent (25%) of the total possible points for said group shall result in:

- a) the application of a fine corresponding to five thousandth percent (0.005%) of the CONTRACT VALUE;
- b) the application of a fine corresponding to five hundredth percent (0.05%) of the CONTRACT VALUE if a result of less than the limit of 25% is achieved in two (2) or more indicators or for two (2) consecutive months, even if in different indicators.
- c) the termination of the CONTRACT if the result of less than 25% is achieved for six (6) consecutive months, even if in different indicators.

Source: Public consultation notice (2013). Appendix IV, pgs. 3-4 and Appendix IV pgs. 12-13. Available at: <http://www.ppp.mg.gov.br/projetos-ppp/projetos-em-elaboracao/rota-lund-1/Edital%20e%20Anexos.zip/view>. Visited on August 9, 2013.

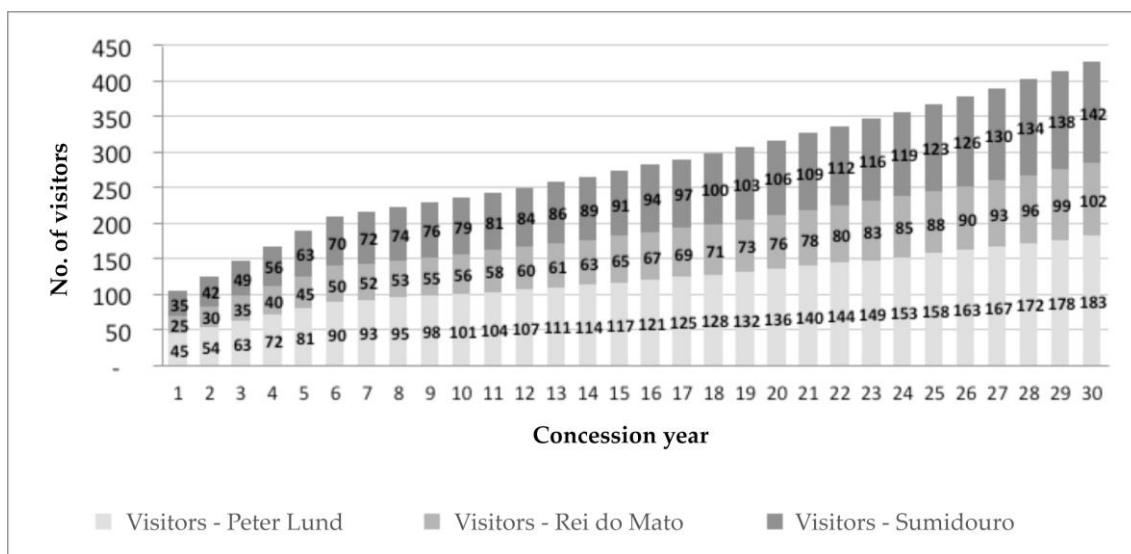
Appendix 8 – Variation in the monthly fee as a function of the Final Performance Score in the Impact Indicators for visitation of 249,000 people /year.



Note: For analysis purposes, considering “estimated average visitation” of 249,000 people per year. Also considered were the annual fee as the simple sum of 12 monthly installments, not considering seasonality, the payment of the annual adjustment and the possibility of the early termination of the contract due to performance scores of less than 0.25 for more than six months (Appendix IV, pgs. 12-13).

Source: the authors.

Appendix 9 - Projected Visitation of the Lund Route UCs ('000 people/year)



Source: Public consultation notice (2013). Appendix XI, pgs. 11-12. Available at: <http://www.ppp.mg.gov.br/projetos-ppp/projetos-em-elaboracao/rota-lund-1/Edital%20e%20Anexos.zip/view>. Visited on August 15, 2013.

Appendix 10 – Variation in fee (Governmental transfer) due to Variation in Ancillary Revenue

Example 1 – Low Visitation Scenario – 105,000 visitors

NFm: Monthly Final Performance Score	1
CP: Gross Consideration (year)	R\$ 6,000,000.00
Rm: Revenue from Ancillary Activities (month)	R\$ 875,000.00
R: revenue from ancillary activities in the period (in this case, annual);	R\$10,500,000.00
P: Minimum government interest in revenue from ancillary activities	2%
Number of visitors (year)	105,000
Km: adjustment factor as a function of the number of visitors: visitors $(10,500/12) \geq (100,000 / 12)$; then $K = \text{Minimum } [10\%; 5.0\% * (10,500 - (100,000/12)) / (100,000/12)]$; therefore, in this case Km = 0.0025	
Mm: adjustment factor as a function of revenue: $(R\$10.5 \text{ million}) \geq (R\$ 50 \text{ million}/12)$; then $M = \text{Minimum } [4\%; 2\% * (10,500 - R\$ 50 \text{ million}/12) / (R\$ 50 \text{ million}/12)]$; therefore, in this case: Mm = 0.03	
Nm: Number of visitors in the period (month);	8,750
T: fee paid per visitor	R\$ 3.00
Average ticket	R\$ 100

$$PM = (6,000,000/12)*1 - [875,000*(0.02+0.0025+0.03)+8750*3] = R\$427,463.50$$

Example 2 – Scenario of High Visitation – 550,000 visitors

NFm: Monthly Final Performance Score	1
CP: Gross Consideration (year)	R\$ 6,000,000.00
Rm: Revenue from Ancillary Activities (month)	R\$ 4,583,000.00
R: revenue from ancillary activities in the period (in this case, annual);	R\$55,000,000.00
P: Minimum government interest in revenue from ancillary activities	2%
Number of visitors (year)	550,000
Km: adjustment factor as a function of the number of visitors: visitors $(550,000/12) \geq (100,000 / 12)$; then $K = \text{Minimum } [10\%; 5.0\% * (550,000 - (100,000/12)) / (100,000/12)]$; therefore, in this case Km = 0.1	
Mm: adjustment factor as a function of revenue: $(R\$ 55 \text{ million}) \geq (R\$ 50 \text{ million}/12)$; then $M = \text{Minimum } [4\%; 2\% * (55,000 - R\$ 50 \text{ million}/12) / (R\$ 50 \text{ million}/12)]$; therefore, in this case: Mm = 0.04	
Nm: Number of visitors in the period (month);	45,833
T: fee paid per visitor	R\$ 3.00
Average ticket	R\$ 100
PM = $(6,000,000/12)*1 - [4,583*(0.02+0.1+0.04)+45,833*3] = -R\\$370,833.00$	

Note: For the purpose of simplification, we considered “R” = Annual Revenue, where $R = 12 * R_m$.

Source: the authors.

Appendix 11 – Cash Flow from the Project

Cash Flow Projection (R\$ '000)	1	2	3	4	5	6	7	8	9	10	15	20	25	30
Year														
Total visitors	105.000	126.000	147.000	168.000	189.000	210.000	216.300	222.789	229.473	236.357	274.002	317.644	368.236	426.887
Average ticket	R\$ 98,44	R\$ 98,21	R\$ 98,04	R\$ 98,36	R\$ 98,80	R\$ 99,43	R\$ 100,53	R\$ 102,13	R\$ 103,55	R\$ 103,36	R\$ 102,49	R\$ 101,74	R\$ 101,09	R\$ 100,53
Revenues	R\$ 20.457.769	R\$ 22.256.338	R\$ 24.012.123	R\$ 25.796.073	R\$ 27.570.747	R\$ 29.352.304	R\$ 30.064.412	R\$ 30.901.476	R\$ 31.729.219	R\$ 32.235.757	R\$ 34.924.399	R\$ 38.154.285	R\$ 42.321.791	R\$ 47.153.072
Net fee	R\$ 10.121.087	R\$ 9.882.320	R\$ 9.600.769	R\$ 9.272.383	R\$ 8.897.220	R\$ 8.472.692	R\$ 8.320.474	R\$ 8.148.313	R\$ 7.966.696	R\$ 7.805.358	R\$ 6.841.782	R\$ 5.837.747	R\$ 5.096.976	R\$ 4.238.220
Income tax	R\$ 2.915.232	R\$ 3.171.528	R\$ 3.421.727	R\$ 3.675.940	R\$ 3.928.831	R\$ 4.182.703	R\$ 4.284.179	R\$ 4.403.460	R\$ 4.521.414	R\$ 4.593.595	R\$ 4.976.727	R\$ 5.436.986	R\$ 6.030.855	R\$ 6.719.313
Net Revenue	R\$ 17.542.537	R\$ 19.084.810	R\$ 20.590.395	R\$ 22.120.133	R\$ 23.641.915	R\$ 25.169.601	R\$ 25.780.234	R\$ 26.498.016	R\$ 27.207.805	R\$ 27.642.162	R\$ 29.947.672	R\$ 32.717.300	R\$ 36.290.935	R\$ 40.433.759
Operating costs	R\$ 18.413.438	R\$ 20.895.523	R\$ 21.061.296	R\$ 20.810.675	R\$ 21.580.897	R\$ 22.351.870	R\$ 22.636.984	R\$ 22.929.120	R\$ 23.228.473	R\$ 23.535.245	R\$ 25.187.843	R\$ 27.059.656	R\$ 29.183.357	R\$ 31.596.706
EBITDA	-R\$ 870.902	-R\$ 1.810.713	-R\$ 470.901	R\$ 1.309.458	R\$ 2.061.018	R\$ 2.817.731	R\$ 3.143.250	R\$ 3.568.896	R\$ 3.979.332	R\$ 4.106.916	R\$ 4.759.830	R\$ 5.657.643	R\$ 7.107.578	R\$ 8.837.053
EBITDA Margin	-R\$ 0	-R\$ 0	-R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0
Depreciation	-R\$ 1.269.069	-R\$ 2.051.581	-R\$ 1.785.639	-R\$ 1.621.658	-R\$ 1.621.658	-R\$ 1.479.673	-R\$ 1.334.292	-R\$ 1.314.636	-R\$ 1.314.636	-R\$ 1.314.636	-R\$ 688.549	-R\$ 730.563	-R\$ 793.584	-R\$ 399.736
Interest	R\$ 0	R\$ 0	R\$ 0	-R\$ 1.110.257	-R\$ 1.083.822	-R\$ 1.057.388	-R\$ 1.030.953	-R\$ 1.004.518	-R\$ 978.084	-R\$ 951.649	-R\$ 819.475	-R\$ 687.302	R\$ 0	R\$ 0
Income before income tax	-R\$ 2.139.971	-R\$ 3.862.295	-R\$ 2.256.540	-R\$ 1.422.457	-R\$ 644.463	R\$ 280.670	R\$ 778.005	R\$ 1.249.741	R\$ 1.686.612	R\$ 1.840.631	R\$ 3.251.805	R\$ 4.239.778	R\$ 6.313.994	R\$ 8.437.318
Income tax	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 66.800	R\$ 185.165	R\$ 297.438	R\$ 401.414	R\$ 438.070	R\$ 773.930	R\$ 1.441.525	R\$ 2.146.758	R\$ 2.868.688
Net Income	-R\$ 2.139.971	-R\$ 3.862.295	-R\$ 2.256.540	-R\$ 1.422.457	-R\$ 644.463	R\$ 213.871	R\$ 592.840	R\$ 952.303	R\$ 1.285.198	R\$ 1.402.561	R\$ 2.477.876	R\$ 2.798.254	R\$ 4.167.236	R\$ 5.568.630
Investment/Reinvestment	-R\$ 20.774.305	-R\$ 18.896.000	R\$ 0	R\$ 0	R\$ 0	-R\$ 1.728.443	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0	R\$ 0
Free cash flow	-R\$ 11.258.054	-R\$ 20.706.713	-R\$ 470.901	R\$ 199.201	R\$ 977.196	-R\$ 34.899	R\$ 1.927.132	R\$ 2.266.939	R\$ 2.599.835	R\$ 2.717.197	R\$ 3.166.425	R\$ 3.528.817	R\$ 4.960.820	R\$ 5.968.365
Discount rate	R\$ 1	R\$ 1	R\$ 1	R\$ 1	R\$ 2	R\$ 2	R\$ 2	R\$ 2	R\$ 2	R\$ 2	R\$ 4	R\$ 6	R\$ 10	R\$ 16
Discounted cash flow	-R\$ 11.258.054	-R\$ 18.824.285	-R\$ 389.174	R\$ 149.663	R\$ 667.438	-R\$ 21.669	R\$ 1.087.816	R\$ 1.163.298	R\$ 1.212.842	R\$ 1.152.357	R\$ 833.819	R\$ 576.990	R\$ 503.650	R\$ 376.242
Free Cash Flow – YTD	-R\$ 11.258.054	-R\$ 31.964.767	-R\$ 32.435.668	-R\$ 32.236.467	-R\$ 31.259.271	-R\$ 31.294.170	-R\$ 29.367.038	-R\$ 27.100.099	-R\$ 24.500.264	-R\$ 21.783.067	-R\$ 8.809.583	R\$ 6.965.837	R\$ 28.072.782	R\$ 53.908.470
Cash flow from ancillary rev EMG														
Nominal	R\$ 547.575	R\$ 786.343	R\$ 1.067.894	R\$ 1.396.279	R\$ 1.771.442	R\$ 2.195.971	R\$ 2.348.189	R\$ 2.520.349	R\$ 2.701.967	R\$ 2.863.305	R\$ 3.826.880	R\$ 4.830.916	R\$ 5.571.687	R\$ 6.430.442
Discounted	R\$ 547.575	R\$ 714.857	R\$ 882.557	R\$ 1.049.045	R\$ 1.209.919	R\$ 1.363.525	R\$ 1.325.491	R\$ 1.293.338	R\$ 1.260.488	R\$ 1.214.321	R\$ 1.007.737	R\$ 789.893	R\$ 565.669	R\$ 405.371

Source: Public consultation notice (2013). Appendix XI, pg. 25. Available at: <<http://www.ppp.mg.gov.br/projetos-ppp/projetos-em-elaboracao/rota-lund-1/Edital%20e%20Anexos.zip/view>>. Visited on August 15, 2013.